

Linux Task 03: Process Management, System Monitoring & Basic Shell Scripting

Part A: Process Monitoring

1. What is a Process?

A process is a program or application that is currently running on a computer. Whenever a user opens an application or executes a command, Linux creates a process to perform that task. Each process uses system resources such as CPU and memory while it is running.

2. What is a PID?

PID stands for Process ID. It is a unique number assigned to every running process in Linux. The operating system uses the PID to identify, monitor, and manage individual processes.

3. Which process is consuming the most CPU?

The process consuming the highest percentage in the **%CPU** column of the top or htop output is using the most CPU resources.

4. Which process is consuming the most Memory?

The process with the highest value in the memory column of top or htop is consuming the most memory.

```
(kali@kali)~$ ps
  PID TTY          TIME CMD
 1786 pts/0    00:00:00 zsh
 2331 pts/0    00:00:00 ps

(kali@kali)~$ ps aux
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.5  0.6 23712 14060 ?        Ss   00:08   0:01 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    00:08   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    00:08   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-kvfree_rcu_reclaim]
root         5  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-rcu_gp]
root         6  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-sync_wq]
root         7  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-netns]
root         9  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/0:0-events]
root        10  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/0:1-cgroup_destroy]
root        11  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/0:0H-events_highpri]
root        12  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/u8:0-ipv6_addrconf]
root        13  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0      0     0 ?        I    00:08   0:00 [rcu_tasks_kthread]
root        15  0.0  0.0      0     0 ?        I    00:08   0:00 [rcu_tasks_rude_kthread]
root        16  0.0  0.0      0     0 ?        I    00:08   0:00 [rcu_tasks_trace_kthread]
root        17  0.0  0.0      0     0 ?        S    00:08   0:00 [ksoftirqd/0]
root        18  0.0  0.0      0     0 ?        R    00:08   0:00 [rcu_preempt]
root        19  0.0  0.0      0     0 ?        S    00:08   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        20  0.0  0.0      0     0 ?        S    00:08   0:00 [rcu_exp_gp_kthread_worker]
root        21  0.0  0.0      0     0 ?        S    00:08   0:00 [migration/0]
root        22  0.0  0.0      0     0 ?        S    00:08   0:00 [idle_inject/0]
root        23  0.0  0.0      0     0 ?        S    00:08   0:00 [cpuhp/0]
root        24  0.0  0.0      0     0 ?        S    00:08   0:00 [cpuhp/1]
root        25  0.0  0.0      0     0 ?        S    00:08   0:00 [idle_inject/1]
root        26  0.0  0.0      0     0 ?        S    00:08   0:00 [migration/1]
root        27  0.0  0.0      0     0 ?        S    00:08   0:00 [ksoftirqd/1]
root        28  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/1:0-rcu_gp]
root        29  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/1:0H-kblockd]
root        30  0.4  0.0      0     0 ?        I    00:08   0:01 [kworker/u9:0-events_unbound]
root        31  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/u10:0-flush-8:0]
root        33  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/u10:1-events_unbound]
root        34  0.0  0.0      0     0 ?        S    00:08   0:00 [kdevtmpfs]
root        35  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-inet_frag_wq]
root        36  0.0  0.0      0     0 ?        S    00:08   0:00 [kauditd]
root        37  0.0  0.0      0     0 ?        S    00:08   0:00 [khungtaskd]
root        38  0.0  0.0      0     0 ?        S    00:08   0:00 [oom_reaper]
root        39  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/u9:1-events_unbound]
root        40  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/R-writeback]
root        41  0.0  0.0      0     0 ?        I    00:08   0:00 [kworker/u9:2-events_unbound]
root        42  0.0  0.0      0     0 ?        S    00:08   0:00 [kcompactd0]
root        43  0.0  0.0      0     0 ?        SN   00:08   0:00 [ksmd]
```

```
(kali@kali)-[~]
$ top
top - 00:16:55 up 8 min, 1 user, load average: 0.50, 0.34, 0.19
Tasks: 181 total, 1 running, 180 sleeping, 0 stopped, 0 zombie
%Cpu(s): 9.7 us, 20.1 sy, 0.0 ni, 70.0 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 1973.7 total, 695.0 free, 908.2 used, 529.3 buff/cache
MiB Swap: 953.7 total, 953.7 free, 0.0 used, 1065.4 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
  769 root        20   0 428652 145132 63372 S 34.3   7.2   0:18.16 Xorg
4260 kali        20   0 271544 28368 21272 S 11.6   1.4   0:00.52 xfce4-screensho
1277 kali        20   0 893144 131420 82780 S  9.6   6.5   0:03.61 xfwm4
1098 kali         9 -11 112468 15564 10704 S  1.7   0.8   0:00.28 pipewire
1206 kali        20   0 215384 3156 2860 S  1.0   0.2   0:02.34 VBoxClient
1343 kali        20   0 296164 48556 20812 S  0.7   2.4   0:02.07 wrapper-2.0
1749 kali        20   0 650316 62528 53204 S  0.7   3.1   0:03.57 qterminal
  49 root        20   0      0      0      0 I  0.3   0.0   0:01.04 kworker/1:1-events_freezable_pwr_efficient
 830 mysql        20   0 1522508 154892 29196 S  0.3   7.7   0:01.84 mariadb
1103 kali         9 -11 612808 34048 26368 S  0.3   1.7   0:00.30 wiraplumber
1345 kali        20   0 272364 28292 21188 S  0.3   1.4   0:01.63 wrapper-2.0
4217 root        20   0      0      0      0 I  0.3   0.0   0:00.01 kworker/1:0-events
   1 root        20   0 23712 14060 10564 S  0.0   0.7   0:01.36 systemd
   2 root        20   0      0      0      0 S  0.0   0.0   0:00.01 kthreadd
   3 root        20   0      0      0      0 S  0.0   0.0   0:00.00 pool_workqueue_release
  4 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
  5 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-rcu_gp
  6 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-sync_wq
  7 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-slub_flushwq
  8 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-netns
 10 root        20   0      0      0      0 I  0.0   0.0   0:00.03 kworker/0:1-cgroup_destroy
 11 root        0 -20      0      0      0 I  0.0   0.0   0:00.01 kworker/0:0H-events_highpri
 12 root        20   0      0      0      0 I  0.0   0.0   0:00.00 kworker/u8:0-ipv6_addrconf
 13 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
 14 root        20   0      0      0      0 I  0.0   0.0   0:00.00 rcu_tasks_kthread
 15 root        20   0      0      0      0 I  0.0   0.0   0:00.00 rcu_tasks_rude_kthread
 16 root        20   0      0      0      0 I  0.0   0.0   0:00.00 rcu_tasks_trace_kthread
 17 root        20   0      0      0      0 S  0.0   0.0   0:00.17 ksoftirqd/0
 18 root        20   0      0      0      0 I  0.0   0.0   0:00.31 rcu_preempt
 19 root        20   0      0      0      0 S  0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
 20 root        20   0      0      0      0 S  0.0   0.0   0:00.03 rcu_exp_gp_kthread_worker
 21 root        rt   0      0      0      0 S  0.0   0.0   0:00.03 migration/0
 22 root       -51   0      0      0      0 S  0.0   0.0   0:00.00 idle_inject/0
 23 root        20   0      0      0      0 S  0.0   0.0   0:00.00 cpuhp/0
 24 root        20   0      0      0      0 S  0.0   0.0   0:00.00 cpuhp/1
 25 root       -51   0      0      0      0 S  0.0   0.0   0:00.00 idle_inject/1
 26 root        rt   0      0      0      0 S  0.0   0.0   0:00.20 migration/1
 27 root        20   0      0      0      0 S  0.0   0.0   0:00.24 ksoftirqd/1
```

Part B: Process Management

```
(kali@kali)-[~]
$ ps aux | grep sleep
kali    9976  0.0  0.0   5580  1924 pts/1    S+   00:32   0:00 sleep 300
kali    10198 0.0  0.1   5580  2024 pts/0    S+   00:33   0:00 sleep 300
kali    10281 0.0  0.1   6528  2248 pts/2    S+   00:33   0:00 grep --color=auto sleep

(kali@kali)-[~]
$ kill 9976

(kali@kali)-[~]
$ ps aux | grep sleep
kali    10198 0.0  0.1   5580  2024 pts/0    S+   00:33   0:00 sleep 300
kali    10511 0.0  0.1   6528  2112 pts/2    S+   00:33   0:00 grep --color=auto sleep

(kali@kali)-[~]
$
```

PID Found

The PID found for the sleep process was **9976**

Command Used

kill PID

Result

The process was successfully terminated and no longer appeared in the process list

Part C: System Monitoring

- 1.Total RAM:The total RAM installed on the system is 1.9 GiB.
- 2.Available RAM: The available RAM on the system is 1.1 GiB.
- 3.Disk Usage: The main storage partition (/dev/sda1) has a total size of 79 GB, out of which 17 GB is used and 58 GB is available. The current disk usage is 23%.
- 4.System Uptime: The system has been running for 28 minutes since the last boot.
- 5.Kernel Version: The system is running on Linux Kernel 6.12.38+kali-amd64

```
(kali@kali)-[~]
└─$ free -h
Mem:          total        used        free      shared  buff/cache   available
Swap:        953Mi       0B         953Mi       18M         540Mi         1.1Gi

(kali@kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            921M   0  921M   0% /dev
tmpfs           198M  988K  197M   1% /run
/dev/sda1       79G   17G   58G  23% /
tmpfs           987M  4.0K  987M   1% /dev/shm
tmpfs           5.0M   0  5.0M   0% /run/lock
tmpfs           1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           987M  344K  987M   1% /tmp
tmpfs           1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  120K  198M   1% /run/user/1000

(kali@kali)-[~]
└─$ uptime
00:37:30 up 28 min,  1 user,  load average: 0.25, 0.27, 0.21

(kali@kali)-[~]
└─$ uname -a
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64 GNU/Linux
```

System Summary Report

The system resources were monitored using Linux system monitoring commands. The free -h command showed that the system has 1.9 GiB of RAM, with 1.1 GiB currently available. Disk usage was checked using the df -h command, which showed that the main storage partition has 79 GB of total space, with 17 GB used and 58 GB available.

The uptime command indicated that the system has been running continuously for 28 minutes. The uname -a command provided information about the operating system and kernel version, which is Linux Kernel 6.12.38+kali-amd64.

Overall, the system is operating normally with sufficient available memory and disk space. Regular monitoring of these resources helps maintain system performance, stability, and security

Part D: Service Monitoring

```
(kali@kali)-[~]
└─$ systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)

(kali@kali)-[~]
└─$ systemctl status NetworkManager
● NetworkManager.service - Network Manager
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-15 00:08:56 EDT; 33min ago
 Invocation: db6d2e7a5cdf43878591bba5e78ad53
     Docs: man:NetworkManager(8)
   Main PID: 627 (NetworkManager)
    Tasks: 4 (limit: 2208)
  Memory: 14.7M (peak: 15.2M)
     CPU: 480ms
   CGroup: /system.slice/NetworkManager.service
           └─627 /usr/sbin/NetworkManager --no-daemon

Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5107] policy: set 'Wired connection 1' (eth0) as default for IPv4 routing and DNS
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5624] device (eth0): state change: ip-config → ip-check (reason 'none', managed-type: 'full')
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5828] device (eth0): state change: ip-check → secondaries (reason 'none', managed-type: 'full')
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5830] device (eth0): state change: secondaries → activated (reason 'none', managed-type: 'full')
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5856] manager: NetworkManager state is now CONNECTED_SITE
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5859] device (eth0): Activation: successful, device activated.
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5873] manager: NetworkManager state is now CONNECTED_GLOBAL
Jun 15 00:08:56 kali NetworkManager[627]: <info> [1781496536.5899] manager: startup complete
Jun 15 00:11:37 kali NetworkManager[627]: <info> [1781496697.0578] agent-manager: agent[18c90ab7b51a54ae,1.53/org.freedesktop.nm-applet/1000]: agent registered
Jun 15 00:16:08 kali NetworkManager[627]: <info> [1781496968.6263] policy: set 'Wired connection 1' (eth0) as default for IPv6 routing and DNS
```

What is a Service?

A service is a background program that runs continuously on a Linux system and performs specific tasks without requiring direct user interaction. Services usually start automatically when the system boots and continue running in the background.

Why are Services Important?

Services are important because they provide essential system functions such as networking, remote access, printing, and security. Many applications and system features depend on services to operate properly.

How can a stopped service affect a system?

If an important service stops running, the related functionality may become unavailable. For example, if the SSH service stops, users will not be able to connect to the system remotely. Similarly, if network services stop, internet connectivity may be affected.

Part E: Shell Scripting

```
(kali@kali)-[~]
$ ./system_report.sh
System Information Report
User: kali
Hostname: kali
Date & Time: Mon Jun 15 12:54:17 AM EDT 2026
Current Directory: /home/kali

Available Memory:

```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	905Mi	678Mi	18Mi	549Mi	1.0Gi
Swap:	953Mi	0B	953Mi			

```

Disk Usage:

```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	921M	0	921M	0%	/dev
tmpfs	198M	992K	197M	1%	/run
/dev/sda1	79G	17G	58G	23%	/
tmpfs	987M	4.0K	987M	1%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	987M	572K	987M	1%	/tmp
tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service

Part F: Security Monitoring Challenge

1. netstat

Purpose

The netstat command is used to display active network connections, listening ports, routing tables, and network statistics. It helps administrators monitor network activity on a system.

Example Output

```
tcp 0 0 192.168.1.5:22 192.168.1.10:54321 ESTABLISHED
```

Security Use Case

Security analysts use netstat to identify suspicious network connections, detect unauthorized remote access, and monitor open ports that could be targeted by attackers.

2. ss

Purpose

The ss command is a modern replacement for netstat. It displays information about network sockets and active connections more efficiently.

Example Output

```
tcp ESTAB 0 0 192.168.1.5:ssh 192.168.1.10:54321
```

Security Use Case

The ss command helps identify active network sessions and detect unusual connections that may indicate malicious activity or unauthorized access attempts.

3. who

Purpose

The who command displays information about users currently logged into the system.

Example Output

```
kali pts/0 2026-06-15 12:30
```

Security Use Case

Administrators can use the who command to detect unauthorized logins and verify which users are currently accessing the system.

4. w

Purpose

The w command shows currently logged-in users and provides additional details such as login time, idle time, and current activity.

Example Output

USER	TTY	FROM	LOGIN@	IDLE	JCPU	PCPU	WHAT
kali	pts/0	12:30	2:00	0.05s	0.02s		bash

Security Use Case

The w command helps administrators monitor user activity and identify unusual behavior that could indicate a compromised account.

5. last

Purpose

The last command displays the login history of users by reading system log records.

Example Output

```
kali pts/0 192.168.1.10 Mon Jun 15 12:30
```

Security Use Case

Security analysts use the last command to investigate login activity, detect unauthorized access attempts, and review historical user sessions during security investigations.

Part G: Mini SOC Activity

1. How would you identify resource-heavy processes?

I would use commands such as top, htop, and ps aux to monitor running processes and their resource usage. These commands display CPU and memory consumption, making it easy to identify processes that are using excessive system resources and affecting performance

2. How would you determine whether a process is suspicious?

To determine whether a process is suspicious, I would examine its name, PID, owner, resource usage, and behavior. Processes with unusual names, unexpectedly high CPU or memory usage, unknown locations, or unauthorized network connections may indicate suspicious activity and should be investigated further

3. What information would you collect before terminating a process?

Before terminating a process, I would collect important information such as the Process ID (PID), process name, user account running the process, CPU and memory usage, and any related network connections. I would also verify whether the process is a legitimate system or application process to avoid accidentally stopping an essential service.